UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Strategic Management Accounting

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The Balanced Scorecard

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Learning Objectives

1. Give examples of measures that fit each of the four scorecard perspectives.
2. Name the four types of quality cost and use them to prepare a quality cost report.
3. Calculate throughput time, delivery cycle time, manufacturing cycle efficiency and overall equipment effectiveness.
4. Construct a balanced scorecard, link its measures and use it to evaluate a strategy.



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Strategy comes first

Section I

The Balanced Scorecard

Why profit alone is not enough

It arrives late

Profit tells us how the last quarter went. By the time it falls, the causes happened months ago.

It does not say why

A lower margin may come from defects, slow deliveries, lost customers or untrained staff. The number alone does not tell us which.

It can be managed

Cutting training or maintenance raises profit this year and damages the business next year.

Financial results remain the final test. The point is that they need measures that show what is happening **before** the money moves.

Two reasons for non-financial measures

REASON 1

They capture fundamental changes

Non-financial and operational indicators measure the fundamental changes a company is making.

Financial measures are **lag** indicators: they summarise the results of past actions.

Non-financial measures are **leading** indicators of future financial performance.

REASON 2

They are easier to understand and influence

Top managers are usually responsible for financial measures, not lower level managers.

Non-financial measures are more likely to be understood and controlled by lower level managers.

A warehouse supervisor cannot move return on equity, but can move the percentage of orders shipped on time.

Leading and lagging indicators

- A **lagging indicator** records an outcome that has already happened: profit, sales, market share.
- A **leading indicator** moves earlier and helps predict the outcome: training hours, defect rates, delivery times.
- A good scorecard mixes both, so managers can act on the drivers and still check the results.

Leading	Lagging
Hours of staff training	Errors per order
Errors per order	Customer satisfaction
Customer satisfaction	Repeat purchases
Repeat purchases	Profit

Notice: the same measure can be lagging for one step and leading for the next.

DEFINITION

A balanced scorecard is an integrated set of performance measures that are derived from and support a company's strategy.

Integrated means the measures are linked to each other, not just a list.

Derived from means we start from the strategy and then choose the measures.

Support means the measures help people put the strategy into action.

What do we mean by strategy?

- A **strategy** is a company's "game plan" for attracting customers by distinguishing itself from competitors.
- The distinguishing features are called the **customer value proposition**: the reason a customer should choose us and not someone else.
- The value proposition is the essence of strategy. If we cannot state it in one sentence, we are not ready to build a scorecard.

Three value propositions

Customer intimacy

"Choose us because we can customise our products and services to your individual needs better than our competitors".

Operational excellence

"Choose us because we deliver faster, more conveniently and at a lower price than our competitors".

Product leadership

"Choose us because we offer more innovative and higher quality products than our competitors".

Source: Treacy and Wiersema, "Customer Intimacy and Other Value Disciplines", Harvard Business Review, 1993, vol. 71, no. 1

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The framework

Section II

The Balanced Scorecard

Strategy on the left, measures on the right

Vision and strategy

What are our financial goals?

Which customers do we want to serve, and how will we win and keep them?

Which internal business processes are critical to delivering value to customers?



The four perspectives form a chain



Read it from the bottom.

Employees need to keep learning in order to improve internal business processes.

Improving processes is necessary to improve customer satisfaction.

Improving customer satisfaction is necessary to improve financial results.

Every arrow is a hypothesis we can test

If we improve this measure, then that other measure should also improve.

Example

If average training hours per employee go up, then the number of errors per order should go down.

Test

After some months, did errors actually fall? If not, either the training is poor or errors have a different cause.

A strategy map puts the chain on one page

A **strategy map** is a diagram of the strategy:

- the **objectives** in each of the four perspectives
- **arrows** showing which objective should cause which
- everything on **one page**, so each employee sees how their work leads to the results

Then each objective gets a **measure**, a **target** and an **initiative**: that is the scorecard.





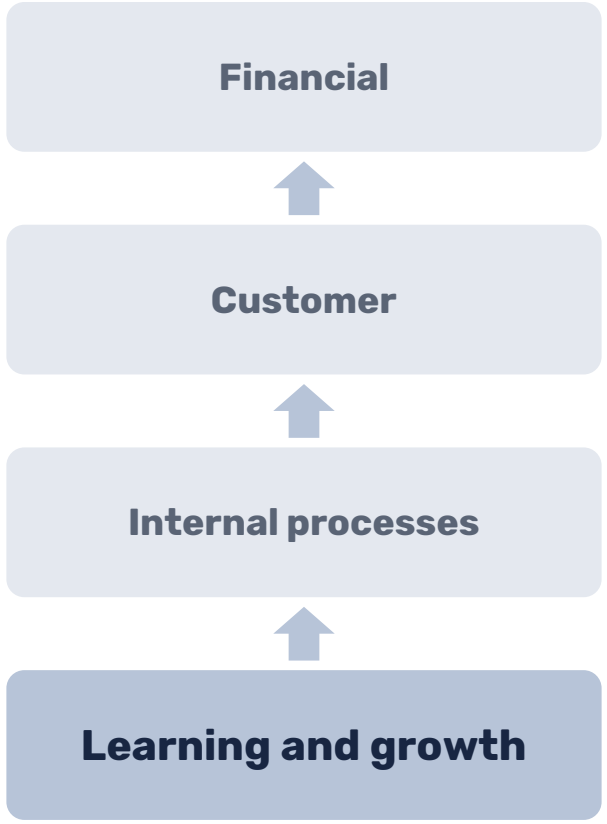
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People, processes and customers

Section III

The Balanced Scorecard

The learning and growth perspective



GUIDING QUESTION

Are we maintaining our ability to change and improve?

What we measure

The skills, motivation, health and stability of the people who run the processes. It is the foundation: everything above depends on it.

Typical measures

Training hours per employee ↑

Employee turnover ↓

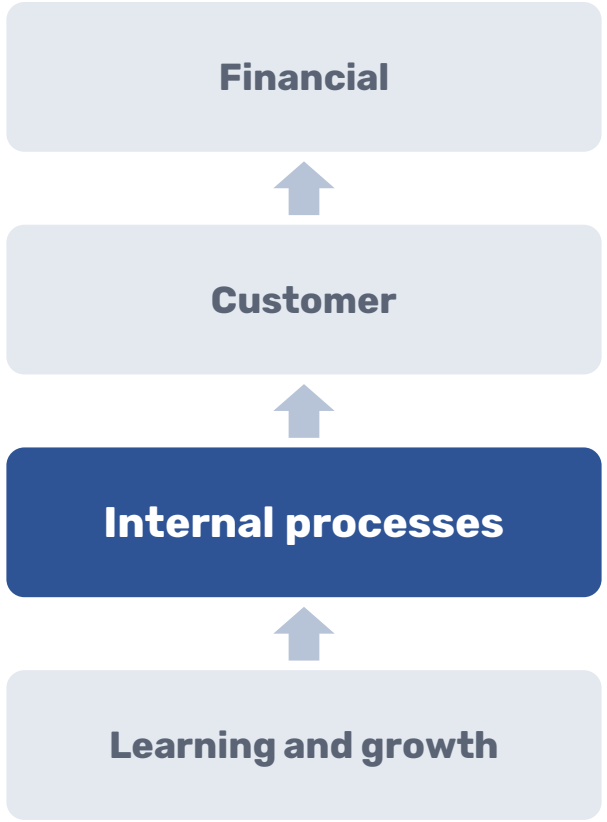
Absenteeism rate ↓

Improvement suggestions per employee ↑

Leadership roles filled internally ↑

Next level up: better trained and more stable people should run **better processes**. These measures move slowly.

The internal process perspective



GUIDING QUESTION

Have we improved key processes so that we deliver more value to customers?

What we measure

How well, how fast and how cheaply the company runs the activities that serve customers, such as assembling a product or handling baggage.

Typical measures

New products designed ↑

Warranty claims ↓

Cost of quality ↓

Delivery cycle time ↓

Manufacturing cycle efficiency ↑

Overall equipment effectiveness ↑

Next level up: better processes should give **customers** something they notice and value.

Six dimensions of process performance

Innovation

How many new products and patents do we create?

Quality

How often do our products and services work right the first time?

Agility

How quickly can we adapt or change?

Cost

How much do we spend on things that add no value?

Time

How long from order to shipment, and how much of it is real work?

Lean resources

Are equipment, inventory and workspace used without waste?

Each strategy looks at its processes with one question

PRODUCT LEADERSHIP

“Is it new and better?”

Watch innovation and quality:

- New products designed ↑
- Patents approved ↑
- Defects ↓

OPERATIONAL EXCELLENCE

“Is it cheap and fast?”

Watch cost and time:

- Cost per unit ↓
- Delivery cycle time ↓
- Waste ↓

CUSTOMER INTIMACY

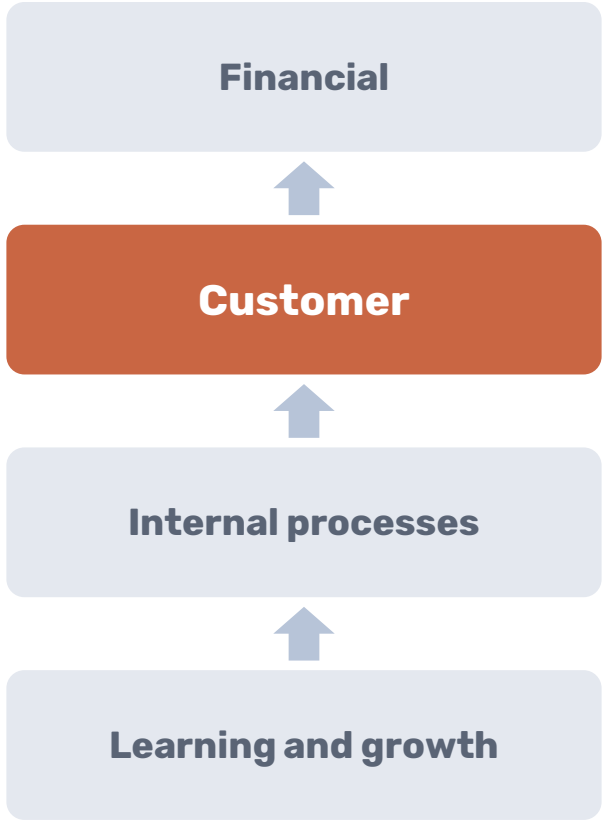
“Is it made for me?”

Watch service and flexibility:

- Complaints solved at first contact ↑
- Custom orders on time ↑
- Small batch sizes ↓

Every company watches all three, but the strategy decides **which one comes first** on a short scorecard.

The customer perspective



GUIDING QUESTION

Do customers recognise that we are delivering more value?

What we measure

Whether customers notice our improvements and change behaviour: they buy again, buy more and recommend us. The key word is recognise.

Typical measures

Market share ↑

Customer defection rate ↓

Customer referrals ↑

Share of wallet ↑

Orders delivered on time, as customers see it ↑

Next level up: customers who buy more and stay longer should produce **better financial results**.



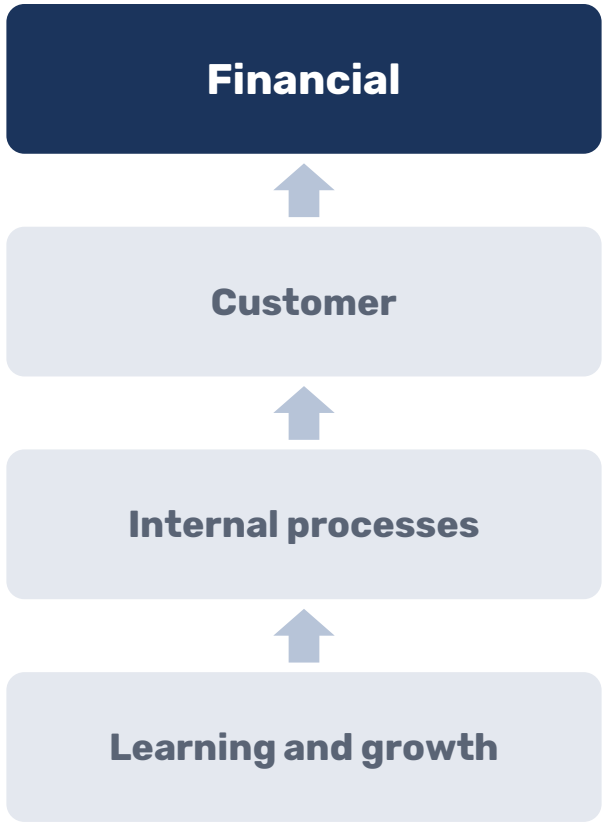
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Financial measures

Section IV

The Balanced Scorecard

The financial perspective



GUIDING QUESTION

Has our financial performance improved?

What we measure

The results shareholders care about. It is the final check: if better people, processes and customers never show up here, the strategy itself may be wrong.

Typical measures

Sales growth rate ↑

Net operating income ↑

Residual income ↑

Return on assets ↑

Net cash flow from operations ↑

Earnings per share ↑

Ratios in the financial perspective

Managers compute many ratios for management purposes. In the balanced scorecard they appear under three headings:

Profitability ratios

- Gross margin percentage
- Net profit margin percentage
- Return on assets (ROA)
- Return on equity (ROE)

Profits

- Contribution margin ↑
- Net operating income ↑
- Residual income ↑

Market performance

- Stock price ↑
- Earnings per share (EPS) ↑
- Price earnings ratio ↑



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The cost of quality

Section V

The Balanced Scorecard

What are quality costs?

Costs incurred to **prevent** defects, or that **result from** defects in products, are known as quality costs.

Many companies are working hard to reduce them

Quality costs are often large and spread across many departments.

Those that succeed have high quality of conformance

The overwhelming majority of their products conform to design specifications and are free from defects.

Four types of quality cost

BEFORE DEFECTS

Prevention costs

Support activities whose purpose is to stop defects from occurring at all.

BECAUSE OF DEFECTS

Internal failure costs

Result from defects found **before** the product is shipped: scrap, rework, downtime.

BEFORE DEFECTS

Appraisal costs

Inspection costs: spent to find defective products before they reach customers.

BECAUSE OF DEFECTS

External failure costs

Arise when a defective product reaches the customer: warranty, recalls, lost sales.

Cost of quality = prevention + appraisal + internal failure + external failure. The goal is to lower this total over time.

Spend a little more before, lose much less after

INVEST

Prevention ↑ and appraisal ↑



SAVE

Internal failure ↓ and external failure ↓

- An effective quality programme makes extra investments in prevention and appraisal that are **more than offset** by lower failure costs, so the total cost of quality falls.
- Over time, appraisal should also fall: when defects are prevented at the source, there is less to inspect.

THE RULE TO REMEMBER

Preventing defects is better than detecting them after they have occurred.

Every euro moved from failures to prevention is a sign that the quality system is maturing.



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Quality cost reports

Section VI

The Balanced Scorecard

What a quality cost report shows

- A **quality cost report** details prevention costs, appraisal costs and the costs of internal and external failures arising from the company's current quality efforts.
- It is used to monitor progress in lowering the overall cost of quality, usually by comparing two or more periods.

How it is laid out

1. The four categories, each with its detailed lines and a subtotal
2. A grand total: the cost of quality
3. Each amount also shown as a **percentage** of a base, usually total sales
4. One column per year, side by side

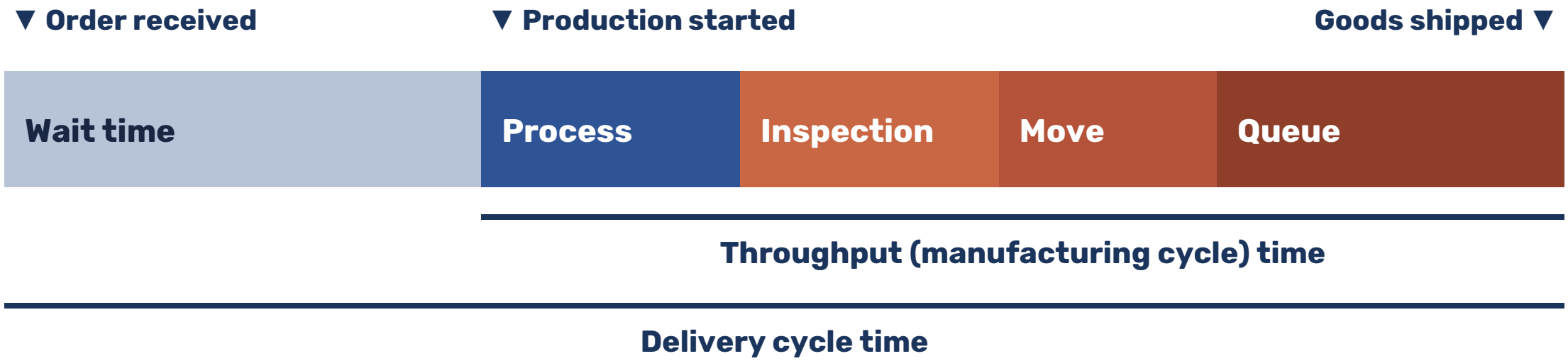
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Time is a performance measure

Section VII

The Balanced Scorecard

The life of a customer order



Value-added

Process time only: when work is actually done on the product.

Non-value-added

Wait, inspection, move and queue time. Reduce them as much as possible.

Throughput time (manufacturing cycle time)

The elapsed time from when production is started until finished goods are shipped to customers. Goal ↓.

Throughput time = Process + Inspection + Move + Queue

Process

Time when work is actually done on the product.

Inspection

Time spent making sure the product is not defective.

Move

Time to carry materials or partly finished products between workstations.

Queue

Time waiting to be worked on, moved, inspected or shipped.

Delivery cycle time

The elapsed time from when a customer order is **received** until the finished goods are **shipped**. This is the time the customer actually experiences. Goal ↓.

Delivery cycle time = Wait time + Throughput time

Wait time is the time from receiving the order until production starts. It adds no value. Companies that drastically cut wait time and the non-value-added parts of throughput time can raise both customer satisfaction and profit.

Manufacturing cycle efficiency (MCE)

MCE = Value-added time (process time) ÷ Throughput time

- MCE puts throughput time into perspective. Goal ↑, and the maximum is 1.
- Any non-value-added time makes MCE **less than 1**. An MCE of 0.5 means half the production time is spent inspecting, moving and waiting.

below 0.10

In many manufacturing companies MCE is below 10%: more than 90% of the time a unit is in process adds no value.



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Overall equipment effectiveness

Section VIII

The Balanced Scorecard

Three ways a machine can disappoint us

Overall equipment effectiveness measures the productivity of a piece of equipment along three dimensions. It is widely used by lean manufacturers. Goal ↑.

Utilisation

Is the machine running when it could be? Breakdowns and long setups leave it idle.

Efficiency

When it runs, does it run at full speed? Minor stoppages slow it down.

Quality

Of what it produces, how much is free of defects?

OEE is the product of three rates

OEE = Utilisation rate × Efficiency rate × Quality rate

Rate	Formula
Utilisation rate	Actual run time ÷ Machine time available
Efficiency rate	Actual run rate ÷ Ideal run rate
Quality rate	Defect free output ÷ Total output

Each rate is between 0 and 1, so OEE is always smaller than the weakest of the three. Three rates of 90% give an OEE of only **73%**.



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Building a scorecard

Section IX

The Balanced Scorecard

Four steps to a scorecard

1

Define the strategy

Name the target customers and the value proposition.

2

Choose measures

A few per perspective, each with a direction.

3

Link them in a strategy map

Draw arrows and write each arrow as an if/then statement.

4

Test and reward

Track the data, check the links, and only later tie pay to the measures.

In brief: define the strategy, translate it into measures that track its execution, and link those measures in a **cause and effect** manner.

What makes a good if/then statement

WEAK

“If employees are happier, the company will do better”.

Neither side is a measure, there is no direction, and nothing can be checked.

STRONG

“If employee turnover ↓, then the number of warranty claims ↓”.

Two named measures, a direction for each, and a mechanism we can explain: experienced staff make fewer mistakes.

Checklist: two specific measures · a direction for each · usually one perspective to the next · a plausible reason · data you can actually collect.



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Using the scorecard

Section X

The Balanced Scorecard

Tie pay to the scorecard, but not on day one

- Bonuses can, and probably should, be linked to scorecard measures. Otherwise the measures are likely to be ignored.
- But only after the organisation has been managed with the scorecard for some time, **perhaps a year or more**.

First be sure the measures are

- reliable
- sensible
- understood by the people evaluated
- not easily manipulated

Beyond shareholders

- Companies must produce financial results that satisfy shareholders. They also have a responsibility to other stakeholders whose interests are tied to the company's performance.
- **Corporate social responsibility (CSR)** means considering the needs of all stakeholders when making decisions. It goes beyond legal compliance to include voluntary actions.

Stakeholders

- Customers
- Employees
- Suppliers
- Communities
- Environmental advocates
- Human rights advocates

Three ways to bring CSR into the scorecard

Inside the four boxes

Add social and environmental measures to the existing perspectives.

A fifth perspective

Add a category dedicated to a wider set of stakeholders.

A separate scorecard

Build a dedicated scorecard for the company's own, or its suppliers', social and environmental performance.

Whichever format is chosen, two rules hold: link the measures with if/then statements, and tie them to how senior managers are evaluated and rewarded. **Otherwise a proclaimed commitment may not lead to action.**